Preliminary Report:
Automated Image Captioning in the Grocery Retail Domain via Tag-Constrained Structural Routing

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1. Introduction

1.1 Aims and Objectives

This project proposes a three-stage, data-driven pipeline for automated image captioning in the grocery retail domain. The central research problem is the prevalence of hallucination in Large Vision-Language Models (VLMs). This phenomenon involves the generation of plausible but visually unverified content, which renders current end-to-end generative models unsuitable for operationally critical deployment. Prior tag-guided captioning approaches, such as Tag2Text and Semantic Compositional Networks, have demonstrated that semantic tags can meaningfully guide caption generation [1], [2]. However, none have addressed domain-specific retail challenges or incorporated unsupervised scenario routing as a structural constraint mechanism.

This project decomposes the captioning task into three layers: (1) a Deterministic Tagging Layer employing a hierarchical four-tier ontology, (2) an Unsupervised Routing Layer that autonomously discovers operational scenarios from tag co-occurrence patterns, and (3) a Conditional Captioning Layer using structured prompt injection with Explicit Absence Encoding. These layers are collectively referred to as the Mixture of Prompts (MOP) architecture.

The project is organised around one primary and three subsidiary research questions, each addressed by a specific experimental component.

Primary Research Question (PRQ): Can a data-driven, tag-constrained pipeline, comprising hierarchical deterministic tagging, unsupervised scenario routing, and explicit absence encoding, significantly reduce object hallucination and increase operational actionability in VLM-generated retail image captions compared to an unconstrained end-to-end baseline?

RQ1: Does unsupervised clustering on tag co-occurrence patterns produce semantically valid and operationally coherent retail scenario groupings without manual expert input?

RQ2: Does Explicit Absence Encoding statistically reduce object hallucination rates (CHAIR) and improve object existence accuracy (POPE) relative to tag-injection-only and unconstrained baselines?

RQ3: Does persona-conditioned captioning via the automated routing layer produce captions rated as more operationally actionable (RAS) than flat-tagged or unconstrained alternatives?

1.2 Domain and Dataset

The target domain is grocery retail environments. The dataset has been sourced from HuggingFace, comprising approximately 10,000 grocery retail images across multiple UK retail brands (Tesco, Waitrose, Asda, and others) with campaign-level, brand-level, and category-level organisation encoded in the file-path hierarchy. This rich file-path metadata encoding campaign type (e.g., Halloween2024, FullStores), retail brand, product category, and store location directly informs the Automated Dynamic Dictionary without requiring external data sources.

1.3 Background and Motivations

The automated interpretation of visual data in retail environments has become an increasingly important capability. Applications span shelf audit automation, planogram compliance, accessibility services, and smart product cataloguing. Recent advances in deep learning and vision-language pretraining have produced powerful image captioning systems [3]. Nevertheless, a fundamental tension persists between generative fluency and factual fidelity.

Large Vision-Language Models trained to maximise token-level likelihood are inherently probabilistic. However, retail environments demand the precise identification of product states, fixture types, and operational anomalies. In such strict contexts, the probabilistic nature of VLMs frequently causes hallucination. The models generate descriptions of objects, promotions, or stock conditions that are absent from the image yet statistically plausible based on their training priors [4]. For example, describing an empty shelf as fully stocked or fabricating promotional signage based on learned associations poses an unacceptable risk in operational deployments.

This project addresses this problem through a structural constraint approach. By extracting a verified set of visual tags prior to generation and conditioning the VLM on both confirmed visual evidence and confirmed absences, the model's generative entropy is reduced and hallucination is structurally discouraged.

1.4 Proposed Methodology

The system implements a three-stage pipeline: (1) Dynamic Context Tagging, (2) Unsupervised Routing, and (3) Conditional Captioning. In this sequence, each stage produces a structured output consumed by the next, enabling independent evaluation and targeted ablation.

1.4.1 Stage 1: Dynamic Context Tagging (Deterministic Layer)

Stage 1 produces a verified, structured tag set T for each input image I, including only visual elements that have crossed defined confidence thresholds. Four extraction modules correspond to the four levels of the hierarchical ontology.

L1 (Scene/Zone): CLIP (ViT-L/14) [17] performs zero-shot classification of the overall spatial context using text prompts such as "a photo of a retail trading floor aisle", "a photo of a supermarket checkout zone", and "a photo of a retail stockroom".

L2 (Fixture Category): Grounding DINO [18], an open-vocabulary detection model capable of localising objects from free-text descriptions, identifies planogram fixture types (e.g., Gondola Shelving, Endcap Display, Pallet Stack, Refrigerated Bay) without any bounding-box annotation or domain-specific fine-tuning. This replaces the originally planned YOLOv8 fine-tuning approach, which was determined to be infeasible given the absence of bounding-box annotated training data and the time constraints of an MSc project.

L3 (Product Type): A two-step process is employed. First, the Gemini API [20] generates a probabilistic list of expected contextual keywords from file-path metadata. For example, the path `train/Halloween2024/Tesco/Food - Floral/` produces a dictionary enriched with terms such as Pumpkin, Cobweb, and Witch Hat. CLIP then performs zero-shot matching against this dynamically generated vocabulary. Critically, PP-OCR [19] text extraction results are fed forward into this stage as a visual grounding constraint. The Gemini prompt is instructed to generate product candidates only within the scope of visually confirmed text. This Dual-Verification mechanism was introduced during development after observing that the LLM would otherwise fabricate plausible but non-existent brand names from metadata alone. The extracted tags are subsequently mapped onto the GS1 Global Product Classification (GPC) Brick-level vocabulary, which is the internationally standardised product classification schema used across UK grocery retail supply chains.

L4 (Attribute/State): PP-OCR [19] extracts visible text (price tags, promotional signage), and a CLIP-based zero-shot attribute classifier identifies operational states (Empty Shelf, Damaged Product, On Promotion) via prompt matching against state-description text templates.

Tags are classified into two tiers: Hard Tags and Soft Tags. The initial proposal specified confidence thresholds of 0.85 for Hard Tags and 0.60 for Soft Tags, following standard CLIP zero-shot classification practice. During implementation, empirical calibration on the validation subset revealed that CLIP's zero-shot confidence scores for fine-grained retail product categories were systematically lower than those observed on general-domain benchmarks. Maintaining the original thresholds would have filtered out the majority of valid product tags, producing unacceptably sparse tag sets. The thresholds were therefore recalibrated to Hard ≥ 0.55 and Soft ≥ 0.30, which maximised the F1 score against the human-reviewed ground truth on the validation subset. Tags scanned for but not detected above either threshold are recorded as Confirmed Absences, forming the input to Stage 3's Explicit Absence Encoding mechanism.

1.4.2 Stage 2: Unsupervised Data-Driven Routing (Logic Layer)

Stage 2 transforms the tag set T into an operational Task Context K. This context comprises a Persona (the intended recipient of the caption) and a Primary Goal (the information the caption must convey), derived autonomously from data rather than from manually engineered rules.

Tags at levels L1 (Scene/Zone) and L4 (State/Attribute) are selected as the clustering feature space. Each image is encoded as a multi-hot sparse vector over the combined L1 and L4 tag vocabulary. K-Means clustering is applied, with the optimal K selected by maximising the Silhouette Score [21] across K = 2 to 8. Once clusters are identified, the Gemini API [20] is prompted with each cluster's tag distribution and asked to assign a Persona and Primary Goal, producing an Automated Routing Matrix. For instance, the combination of Trading Floor, Out of Stock, and Price Tag Missing is routed to an Inventory Manager with instructions to identify the out-of-stock product by its shelf adjacency context. Similarly, a combination of Trading Floor, Product Damaged, and Planogram Violation is routed to a Safety Officer to describe the hazard type and severity for incident logging.

1.4.3 Stage 3: Automated Conditional Captioning (Generative Layer)

Stage 3 generates the final natural language caption conditioned on the verified tag set T and the task context K using the Mixture of Prompts (MOP) strategy.

Evolution from Closed-World to Open+Anchor Strategy. The initial proposal described a closed-world constraint model in which the VLM was permitted to mention only objects present in the verified tag set T. During development, this strict constraint was found to eliminate the VLM's ability to describe contextually important visual elements not captured by the L1–L4 ontology, such as seasonal decorations, colour themes, or spatial arrangement patterns that contribute to a complete and natural caption. The prompt strategy was therefore revised to an Open+Anchor approach. The VLM is permitted to freely describe what it observes in the image, but is required to mention all L3-verified products as mandatory anchors. Tags classified as Confirmed Absences remain as explicit negative constraints (e.g., "Do NOT state or guess the following"), and attributes whose confidence falls into an ambiguous range are flagged with instructions for the VLM to refrain from guessing. This revised strategy preserves the VLM's native visual interpretation capability while maintaining the factual grounding guarantees of the MOP architecture.

The Explicit Absence Encoding mechanism converts negative detection outcomes into positive assertions, preventing the VLM from filling unaddressed features with prior-driven fabrications. This acts as a structural extension of the entity-aware prompting approach demonstrated in ViECap [4].

A Priority Rule is enforced within the MOP prompt structure: file-path metadata is confined to providing background semantic context only, while product-level descriptions are strictly locked to visual evidence from L3/L4 tags. This separation was introduced after observing that the VLM would otherwise prioritise metadata over visual evidence. For example, captioning a standard cola product as “Halloween Pumpkin Cola” simply because the file path contained “Halloween2024”.

Stylistic Forcing (Persona Lock-in). An additional prompt-level intervention was introduced to address Instruction Neglect, a behaviour observed in sub-10B parameter VLMs whereby the model progressively disregards system-level instructions as the prompt length increases. In early experiments, the persona assigned by Stage 2 was placed as a system instruction, but the VLM frequently reverted to a generic image-description style within the first sentence. To counteract this, the opening line of every MOP prompt is now hard-anchored with a persona-specific header (e.g., “Merchandising Report:”), forcing the VLM’s initial token generation into the desired register. This stylistic forcing proved effective in maintaining persona-consistent output across extended caption generation.

The generative VLM is currently LLaVA-1.5-7B [22] or llava-phi3 (3.8B), selected for strong instruction-following capability within manageable VRAM requirements on consumer hardware. The choice of a relatively small model is deliberate, as it demonstrates the MOP architecture’s ability to suppress hallucination even in models with limited parametric capacity.

Orchestrator Role. An external LLM API, referred to as the orchestrator, is used across Stages 1 and 2 to perform text-based reasoning tasks that fall outside the scope of local vision models. Specifically, the orchestrator performs three tasks: (1) generating contextual keyword candidates from file-path metadata for the L3 Dynamic Dictionary, (2) mapping extracted product categories onto the GS1 GPC standard vocabulary, and (3) assigning cluster-specific personas and MOP prompt templates from the tag distributions produced by Stage 2’s K-Means clustering. The current orchestrator is the Gemini API [20], selected for its structured JSON output stability and cost efficiency.

1.5 Data Plan

Preprocessing

A Laplacian variance filter rejects images below a sharpness threshold, targeting the retention of at least 9,000 usable images. File-path metadata is parsed and stored as a structured JSON sidecar for each image. The source dataset has been pre-processed to remove personally identifiable information, meaning no additional privacy filtering is applied.

LLM-Assisted Semi-Automated Annotation

As the researcher is not a domain expert, ground truth annotation follows a practical LLM-assisted workflow rather than a pure manual expert process. A stratified sample of 300–400 images is annotated using a two-step approach.

Tag Ground Truth: The researcher directly reviews each sampled image against the L1–L4 ontology schema through visual inspection to produce validated tag labels. The Gemini API is used as a suggestion generator to accelerate this process. Its proposed tags are treated as candidates requiring human confirmation or correction, not as authoritative annotations. A 50-image subset is independently reviewed by a second student annotator to compute inter-rater reliability.

Reference Captions: For a subset of approximately 150–200 images, the Gemini API is prompted to generate two to three candidate reference captions per image. The researcher reviews and lightly edits the most factually accurate candidate. It is explicitly acknowledged that sourcing reference captions from the same model family as the LLM-as-Judge evaluation introduces a potential consistency bias. These metrics are therefore treated as supplementary caption quality indicators only.

1.6 Evaluation Framework

The evaluation framework is structured to answer each research question through a combination of quantitative pipeline metrics, automated evaluation, and small-scale human evaluation.

1.6.1 Stage 1: Tag Detection Quality

Precision and Recall are computed against the annotated ground truth.

1.6.2 Stage 2: Routing Validity (RQ1)

The Silhouette Score [21] and Davies-Bouldin Index are used for cluster cohesion and separation. Routing appropriateness is assessed via researcher review of a small sample per cluster.

1.6.3 Stage 3: Hallucination Reduction (RQ2)

The primary hallucination metric is Retail-CHAIR, a domain-adapted variant of the standard CHAIR metric. Standard CHAIR checks whether object nouns in a generated caption appear in the COCO reference vocabulary. Retail-CHAIR replaces this vocabulary with the project's L1–L4 ontology tags.

A methodological caveat must be noted. Because the MOP architecture structurally requires the VLM to mention verified L3/L4 tags as anchors, the Retail-CHAIR score is expected to be inherently favourable for the proposed system relative to the unconstrained baseline. Retail-CHAIR therefore primarily measures the degree to which the VLM introduces additional object nouns beyond the verified tag set, rather than providing a fully independent assessment of factual accuracy. Consequently, it is complemented by the Pairwise Multimodal LLM-as-a-Judge framework.

Evolution from POPE to Pairwise Multimodal LLM-as-a-Judge. The initial proposal planned to employ POPE as a secondary hallucination metric. During implementation, two practical barriers emerged. Constructing reliable binary probing questions for the diverse L3/L4 vocabulary required extensive manual curation that exceeded available resources. Furthermore, POPE's binary yes/no format could not capture the nuanced differences between MOP and baseline captions in dimensions such as uncertainty handling.

Consequently, POPE has been replaced by a Pairwise Multimodal LLM-as-a-Judge [23] framework. The concrete evaluation protocol operates as follows: (1) the judge model receives the original retail image alongside both the MOP caption and the baseline caption; (2) the two captions are presented with their order randomised per image to prevent position bias; (3) the judge selects the superior caption across Visual Accuracy, Factual Completeness, and Uncertainty Handling; and (4) an overall preference and free-text reasoning are recorded.

This multimodal pairwise design avoids the scale calibration problem inherent in numerical ratings. By providing the actual image to the judge, it enables direct visual verification rather than relying on textual inference alone.

1.6.4 Stage 3: Caption Quality (RQ3)

A Retail Actionability Score (RAS) evaluation will be conducted with a panel of fellow students on a 0–2 scale (0 = Useless, 1 = Informative but non-actionable, 2 = Actionable).

1.6.5 Ablation Study

Three experimental variants will be evaluated. Baseline A employs a VLM with no tag constraints to establish the base hallucination rate. Baseline B uses flat tag injection only, isolating the contribution of routing and absence encoding. The Proposed System tests the full 3-stage pipeline hypothesis.

1.7 Current Limitations and Open Questions

Through implementation, several conceptual limitations have been identified that warrant discussion with the supervisor:

1. **Cascading Error Propagation:** Because the MOP architecture structurally constrains the VLM to the tag set produced in Stage 1, any misclassification by the detection models is forcibly embedded in the final caption. The pipeline is highly effective at preventing the VLM from inventing content, but it cannot correct upstream detection failures.
2. **Inference Latency:** The sequential invocation of multiple vision models and an external API introduces substantially higher latency per image compared to a single end-to-end model.
3. **OCR-to-Product Spatial Association:** PP-OCR extracts text tokens from an image but does not inherently associate each text segment with a specific product or shelf region.
4. **Evaluation Circularity:** Using the Gemini API as both the orchestrator and as the judge evaluator introduces a potential self-preference bias.

1.8 Expected Contributions

This project makes the following anticipated contributions:

Methodological: A novel three-stage data-driven tag-constrained captioning pipeline. It introduces Explicit Absence Encoding and unsupervised tag-to-task routing as structural hallucination mitigation mechanisms not present in prior image captioning literature [1], [2], [4], [5].

Domain: A four-tier retail visual ontology combined with a file-path-metadata-driven Dynamic Dictionary.

Empirical: A comparative ablation study on a real-world grocery retail image dataset demonstrating the individual and combined contributions of each pipeline component.

Practical: A deployable, compute-efficient captioning pipeline with GDPR-compliant anonymisation, suitable for deployment in UK grocery retail contexts.

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2. The Background Material for the Project

The proposed system sits at the intersection of image captioning, semantic tag-guided generation, VLM hallucination mitigation, and domain-specific visual understanding. This section reviews the most relevant prior work and identifies the specific gaps this project addresses.

2.1 Image Captioning: From Statistical to Neural Approaches

Early image captioning systems discovered statistical correlations between image features and descriptive keywords to produce keyword-based annotations of novel images [6]. The field has since been transformed by deep learning, with transformer-based encoder-decoder architectures and large-scale pretraining becoming the dominant paradigm [3]. Comprehensive surveys of deep learning captioning approaches consistently identify hallucination, missing context, and domain generalisation as the primary unresolved challenges in the field [3]. Surveys of earlier approaches further established the evaluation metrics and benchmark datasets, including COCO and Flickr30K, that remain in standard use [7].

2.2 Tag-Guided Image Captioning

The use of image-level semantic tags as generative guidance is well-established. Several early approaches, including Semantic Compositional Networks [2] and semantic attention mechanisms [8], demonstrated that weighting generative parameters with tag probabilities improves overall caption quality. More recent transformer-based architectures, such as HAAV [9] and ViTCAP [5], further confirm that integrating semantic concept tokens directly into the generation pipeline enhances image-text alignment [10] and allows for targeted error correction [11].

Tag2Text [1] is the most directly related prior work to this project. It proposes a generation scheme where a dedicated tag recognition decoder guides a vision-language model's output across general-domain imagery. This project distinguishes itself from Tag2Text on three key dimensions: (1) it uses a hierarchically structured retail ontology to constrain tags to operationally meaningful categories, (2) it introduces an unsupervised routing layer to translate the tag set into an operational task context before generation, and (3) it implements Explicit Absence Encoding to convert confirmed non-detections into positive generative constraints. In a related e-commerce context, Singh [12] validates this kind of attribute-first generation paradigm, showing that embedding predicted product attributes into structured prompts reduces hallucination from 12.7% to 7.1%.

2.3 Domain-Specific and E-Commerce Visual Description Generation

While general-domain captioning research shows the value of tag-guided generation, applying this to commercial and retail domains introduces new challenges. Frameworks like OPAL [13] and TUNA [14] demonstrate that injecting domain-specific schema constraints and retrieval-augmented object tags into MLLM prompts significantly reduces the mention of non-existent objects. Similarly, VRAP [15] shows that offline tag pre-generation can maintain accuracy while reducing inference latency.

While these systems confirm that structured prompt injection provides measurable hallucination suppression, they primarily focus on single-product e-commerce imagery. They do not address the multi-context operational diversity characteristic of grocery retail environments, where a single dataset spans fundamentally different task types such as inventory audits and safety inspections. Furthermore, they do not incorporate unsupervised task-context routing or explicit absence-signal mechanisms as structural constraints.

2.4 Hallucination Mitigation in VLMs

Hallucination in VLMs arises from the tension between pixel-level image evidence and the model's learned inductive bias over plausible image-text associations [4]. ViECap addresses this in zero-shot captioning by constructing entity-aware hard prompts that direct LLM attention toward visually confirmed entities, demonstrating reduced object fabrication [4]. VIVO addresses the related problem of novel object captioning by pretraining on large-scale image-tag pairs using a Hungarian matching loss [16]. While these approaches confirm that structural input constraints can reduce hallucination, none address the specific failure mode of fabricated operational states, such as invented promotional offers, in retail VLM deployment.

2.5 Open-Vocabulary Detection and Scene-Text Recognition

Two technical components central to this project's Stage 1, open-vocabulary object detection and scene-text recognition, draw on recent methodological advances. Grounding DINO [18] marries the DINO detection framework with grounded pre-training to enable open-set object detection from free-text descriptions. This capability is directly exploited in the L2 Fixture Type extraction module, where fixture categories are specified as natural language prompts. PP-OCR [19] provides an ultra-lightweight text detection and recognition pipeline capable of extracting scene text from retail imagery with sub-second latency, forming the backbone of the L4 Attribute extraction module and the Dual-Verification constraint for L3 product tagging.

2.6 Research Gap

The reviewed literature confirms three converging findings: tag-guided captioning improves factual grounding in general domains [1], [2], [5]; retrieved or predicted object tags measurably reduce hallucination when injected as structured prompts [14], [15]; and schema-constrained generation in e-commerce settings produces more accurate outputs than unconstrained end-to-end generation [12], [13].

However, no existing work applies a hierarchically structured retail ontology aligned to GS1 GPC industry standards as the tagging backbone. Furthermore, no existing framework introduces an unsupervised tag co-occurrence routing layer to autonomously assign distinct operational task contexts to images at inference time. Finally, the implementation of Explicit Absence Encoding to convert confirmed non-detections into positive generative constraints remains unexplored as a structural hallucination suppression mechanism. This project addresses all three gaps within a unified, compute-efficient pipeline designed specifically for multi-context grocery retail environments.

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3. Project Schedule

The project runs from late March 2026 to early August 2026 (approximately 18 weeks). The work is organised into eight phases. As of the time of writing, Phases 1 through 7 have been completed, with Phase 8 scheduled to begin immediately.

3.1 Completed Phases

Phase 1: Foundation & Data Acquisition (Weeks 1–2) ✓ Complete

The core environment was established, incorporating CLIP [17], Grounding DINO [18], and PP-OCR [19]. A streaming connection to the HuggingFace dataset was configured, and Laplacian variance-based blur filtering was applied to the images. Furthermore, a file-path metadata parser was implemented to generate structured JSON sidecars for all dataset entries.

Phase 2: L1–L2 Tagging Implementation (Weeks 3–4) ✓ Complete

The CLIP zero-shot L1 scene classification was integrated alongside the Grounding DINO L2 open-vocabulary fixture detection pipeline. Both modules were successfully validated on pilot samples.

Phase 3: GS1 Mapping & L3 Product Tagging (Weeks 4–5) ✓ Complete

Automated Dynamic Dictionary generation was implemented via the Gemini API [20]. CLIP zero-shot L3 product matching was executed against this dynamically generated vocabulary. Extracted product tags were then mapped to the GS1 GPC Brick-level standard. The Dual-Verification mechanism was introduced during this phase after observing LLM hallucination in brand-name generation.

Phase 4: L4 Attribute & OCR Tagging (Weeks 5–6) ✓ Complete

The PP-OCR [19] text extraction module was integrated. A CLIP zero-shot attribute classifier was developed for operational states, including stock levels and shelf conditions. These components were consolidated into a unified hierarchical tag structure.

Phase 5: Unsupervised Routing & Clustering (Week 6) ✓ Complete

K-Means clustering was applied to L1 and L4 co-occurrence vectors, with automated K selection achieved via Silhouette Score [21] maximisation. The Gemini API was subsequently used for persona and goal assignment per cluster, leading to the auto-generation of MOP prompt templates.

Phase 6: MOP Captioning & Baseline (Weeks 6–7) ✓ Complete

LLaVA-1.5-7B [22] and llava-phi3 were integrated via Ollama for local inference. The full MOP prompt assembly pipeline was finalised, incorporating the Open+Anchor strategy, Priority Rule, and Stylistic Forcing. Baseline caption generation was also completed for ablation comparison.

Phase 7: Large-Scale Resilience Refactoring (Week 7) ✓ Complete

The codebase underwent structural refactoring to ensure the full pipeline could process 10,000 images without data loss or redundant computation. Key implementations included dictionary-based checkpointing across all pipeline stages, periodic intermediate state persistence, and automated memory management. All stages were consolidated into a single-command batch execution script for reproducibility.

3.2 Remaining Phases

Phase 8: Full 10K Execution & Evaluation (Weeks 8–13)

Full pipeline execution will commence across all 10,000 images. All three ablation variants will be run on the held-out test set, and Retail-CHAIR metrics will be computed. The Pairwise Multimodal LLM-as-a-Judge [23] evaluation will be conducted alongside the student panel RAS evaluation. A reduced-intensity period is incorporated during the May examination window, during which writing tasks will take priority over compute-intensive evaluation runs.

Dissertation Writing & Submission (Weeks 14–19)

Chapters 4 and 5 (Methodology) will be written in parallel with the Phase 8 evaluation. Chapter 6 (Experiments & Results) will be drafted as evaluation results become available. Chapters 7 and 8 (Discussion and Conclusion) will incorporate supervisor feedback cycles. The final phase includes bibliography formatting, code packaging, and formal submission.

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